

# Sommelier: Scalable Open Multi-turn Audio Pre-processing for Full-duplex Speech Language Models

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As the paradigm of AI shifts from text-based LLMs to Speech Language Models (SLMs), there is a growing demand for full-duplex systems capable of real-time, natural human-computer interaction. However, the development of such models is constrained by the scarcity of high-quality, multi-speaker conversational data, as existing large-scale resources are predominantly single-speaker or limited in volume. Addressing the complex dynamics of natural dialogue, such as overlapping and back-channeling remains a challenge, with standard processing pipelines suffering from diarization errors and ASR hallucinations. To bridge this gap, we present a robust and scalable open-source data processing pipeline designed for full-duplex model.

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Demo and Code: [sommelier.github.io](https://sommelier.github.io)

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## 1 Introduction

Recent advances in speech large language models (SLMs) have evolved from single short queries to multi-turn, open-ended conversations (Xu et al., 2025b; Goel et al., 2025). Yet most systems still operate in disjoint user and assistant turns through a cascaded ASR and TTS pipeline (Ye et al., 2025), which inherits latency, discards paralinguistic cues, and struggles with interruptions, overlapping speech, and backchanneling. Full-duplex system (Wang et al., 2024b; Roy et al., 2026) addresses these limitations by enabling the system to listen and speak simultaneously, supporting more fluid and human-like interaction.

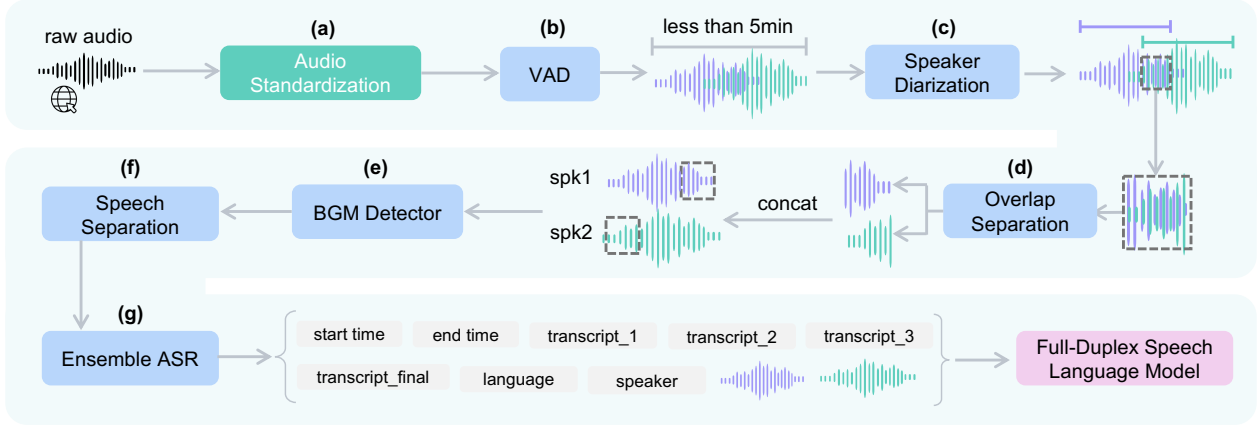
Progress toward full-duplex SLMs has been facing bottlenecked by the lack of high-quality conversational data suitable for duplex training. While Moshi (Défossez et al., 2024) leverages millions of hours of unsupervised audio for pre-training, these sources are largely single-stream and provide limited supervision for overlapping speech. Consequently, overlap robustness relies on relatively small high-fidelity conversational corpora such as Fisher (Cieri et al., 2004), which is unlikely to meet the scale and diversity required for supervised fine-tuning (SFT) (Xu et al., 2025b).

Curating full-duplex training data from in-the-wild recordings is challenging because real conversations contain frequent overlaps, backchannels, and acoustic clutter, which amplify diarization and transcription errors (Wang et al., 2024a). In addition, long-form audio typically includes non-conversational or irrelevant regions (e.g., music, noise, long silences), requiring careful filtering and normalization while preserving speaker structure and multi-turn context. Finally, processing web-scale audio demands high throughput to make large-scale curation feasible under practical compute budgets in the real industry.

To address these challenges, we propose an open, robust, and scalable speech pre-processing pipeline designed for full-duplex SLMs. Our contributions are as follows:

- **The first scalable pipeline for full-duplex SLMs:** We release a scalable pipeline for curating multi-turn conversational speech suitable for full-duplex training, helping alleviate the community-wide data scarcity.
- **High-fidelity overlap processing:** We provide a detailed processing strategy that handles overlaps via rigorous diarization analysis and reduces ASR hallucinations using paralleled model ensembling and n-gram filtering.

♣This work was done during the residency program at NAVER Cloud.



**Figure 1** The overall pipeline of the **Sommelier** conversational audio pre-processing. Blue boxes denote neural model-based components, and a green box represent a algorithmic component.

- **Proven efficacy on a full-duplex model:** We validate our pipeline by fine-tuning the full-duplex model Moshi on **Sommelier**-processed speech and analyze practical data requirements for stable full-duplex training.

## 2 Method

In this section, we present **Sommelier**, a robust data processing pipeline designed to transform raw, in-the-wild conversational audio into high-quality training corpora for full-duplex Speech Language Models (SLMs). Unlike traditional ASR pipelines that prioritize clean, non-overlapping speech, our design philosophy centers on preserving the chaotic yet rich dynamics of human dialogue, such as overlaps and backchannelings, while ensuring scalability for web-scale processing. The overall architecture, illustrated in Figure 1, is built as a modular framework where each component can be toggled or reconfigured, allowing researchers to adapt the trade-off between data purity and conversational authenticity.

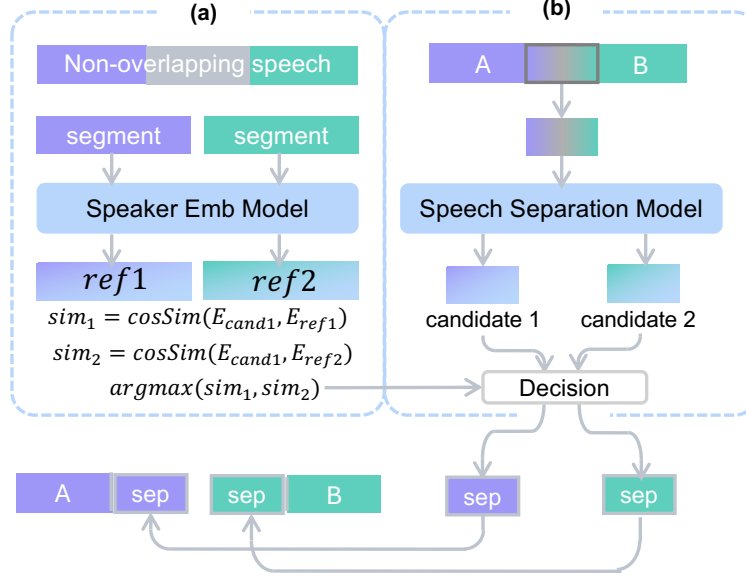
**Sommelier** is designed to transform raw audio into clean, well-structured data while preserving the semantic context. The process begins with standardization, bringing diverse audio formats into a unified representation. We then segment the audio based on silence detection, followed by a Voice Activity Detection (VAD) model (Team, 2024) that further partitions the content into chunks of less than five minutes, a practical constraint that prevents downstream models from running out of memory on lengthy recordings (§ 2.1). Speaker diarization (§ 2.2) follows, identifying who speaks when. Guided by these speaker boundaries, we separate and restore overlapping speech regions (§ 2.3), with optional removal of background noise and music (§ 2.4) depending on the use case. Finally, an ensemble of three Automatic Speech Recognition (ASR) models (§ 2.5) generates text transcripts and captions, leveraging model diversity to improve robustness. Each module in the pipeline can be toggled on or off to suit specific requirements.

Rather than stripping away speech overlaps and backchannelings, interruptions, and simultaneous speech that characterize real dialogue, we preserve them. This allows the duplex speech language model to learn not just what people say, but how conversations actually unfold.

### 2.1 Audio Standardization

Since collected radio and podcast data vary in format and volume, we adopt the method of (He et al., 2024). Using the pydub<sup>1</sup> and librosa (McFee, 2025) libraries, we convert all audio to the standard format (16kHz, 16-bit, Mono) and perform loudness normalization to -20dBFS (He et al., 2024) as illustrated in Figure 1(a).

<sup>1</sup><https://github.com/jiaaro/pydub>



**Figure 2** Illustration of the speech overlap separation process. **(a)** The process of calculating similarity to distinguish speaker identities using arbitrary independent speaker segments. **(b)** Separating overlapped regions and making identity decisions for candidates based on the similarity calculated in (a). Finally, the separated segments are concatenated with the original segments.

## 2.2 VAD & Speaker Diarization

To prevent out-of-memory issues with the diarization model, we split long audio files into units of less than 5 minutes as shown in Figure 1(b). To maintain conversational context, we use a VAD model to cut the audio at silence intervals.

For speaker diarization, as shown in Figure 1(c), instead of the commonly used pyannote speaker-diarization-3.1 model (Bredin, 2023), we adopted *Sortformer* (Park et al.) from NVIDIA. Section 3.2 presents a performance comparison between the two models, demonstrating Sortformer’s superiority in robustly capturing very short utterances such as backchannelings.

## 2.3 Handling Overlapping Speech

Conversational audio features frequent turn changes and short utterances (Wang et al., 2025b). To systematically handle massive-scale industrial speech data, we categorized overlapping scenarios into four distinct cases, as shown in Figure 5. *Case 1* segments based on the overlap, yielding non-overlapping segments but losing full utterance information. *Cases 2* and *Case 3* assign the overlapping speech to one side, which risks ASR errors where utterances mix or transcripts fail. *Case 4* allows both segments to contain the overlap based on speaker identity, preserving full information despite sharing the ASR issues of *Cases 2* and *Case 3*.

We selected *Case 4* as our baseline, incorporating a module that performs two-speaker separation (Shin et al., 2024) on the overlapped intervals. We find that inputting only the duplicated part into separation model works better than using the entire segment.

Before separating overlapped speech, we extract non-overlapping parts longer than 2 seconds to generate embedding tuples  $(e_{ref1}, e_{ref2}) = (\mathcal{M}_{emb}(a_0), \mathcal{M}_{emb}(a_1))$ , where  $a$  denotes the audio segment and  $\mathcal{M}$  represents the model. This process uses the speaker embedding model  $\mathcal{M}_{emb}$  as shown in Figure 2(a). In parallel, the overlapped audio  $a_{overlap}$  is fed into the speech separation model  $\mathcal{M}_{sep}$  to produce candidates  $a_{cand1}$  and  $a_{cand2}$  as shown in Figure 2(b). To identify the speakers, we calculate the cosine similarity scores  $S_1 = \text{sim}(\mathcal{M}_{emb}(a_{cand1}), e_{ref1})$  and  $S_2 = \text{sim}(\mathcal{M}_{emb}(a_{cand1}), e_{ref2})$ . The candidate with the higher similarity is assigned to the corresponding speaker ( $a_i$ ), while the other candidate corresponds to the remaining speaker. Finally, we concatenate the non-overlapping parts with the separated segments to create single-speaker segments as shown in Figure 1(d).

| Model                  | Pause Handling  |              | Backchannel  |              |              | Smooth Turn Taking |           | User Interruption |              |           |
|------------------------|-----------------|--------------|--------------|--------------|--------------|--------------------|-----------|-------------------|--------------|-----------|
|                        | Synthetic TOR ↓ | Candor TOR ↓ | TOR ↓        | Freq ↑       | JSD ↓        | Candor TOR ↑       | Latency ↓ | TOR ↑             | GPT-4o ↑     | Latency ↓ |
| Moshi                  | 0.985           | 0.980        | 1.000        | 0.001        | 0.957        | 0.941              | 0.265     | 1.000             | 0.765        | 0.257     |
| <b>Moshi+Sommelier</b> | 1.000           | 1.000        | <b>0.291</b> | <b>0.052</b> | <b>0.630</b> | <b>1.000</b>       | 0.344     | 0.858             | <b>3.684</b> | 1.065     |

**Table 1** Full-Duplex-Bench 1.0 results for base Moshi and Moshi fine-tuned on 83 hours of **Sommelier**-processed data. Arrows indicate whether higher (↑) or lower (↓) values are better.

## 2.4 Background Music Removal

In addition to multi-speaker overlaps, the diverse nature of industrial audio sources introduces another challenge. Audio from radio broadcasts or dramas contains background music (BGM), which may be undesirable for training speech language models. We employ PANNs (Kong et al., 2020) (Pre-trained Audio Neural Networks) to estimate the probability of background music presence in each segment. If the probability exceeds a threshold of 0.3, we apply the Demucs (Rouard et al., 2023; Défossez, 2021) model to extract the vocal track. Since music removal can degrade speech quality, we selectively apply it only to segments identified by PANNs, minimizing unnecessary processing as shown in Figure 1(e) and (f).

We find that feeding the entire audio context into Demucs yields substantially better separation performance than processing short segments in isolation. Therefore, we input full two-minute audio chunks into the model and subsequently extract only the required portions from the separated output. We also considered SAM-Audio (Shi et al., 2025) for music removal but excluded it due to its high inference latency (RTF 0.73 on A100), which limits its scalability for large datasets.

## 2.5 Ensemble-based ASR

High-quality ASR is essential for constructing large-scale datasets, as it generates the text labels necessary for model training. However, relying on a single model, even SOTA architectures like Whisper (Radford et al., 2022a), poses significant risks. These models are prone to hallucinations, particularly in silent or noisy segments, where they often generate repetitive or nonsensical text (Koencke et al., 2024b; Barański et al., 2025; Mansoor et al., 2025). Such artifacts introduce noise into the training signal, causing the downstream model to mimic these pathological behaviors.

To mitigate this, we employ a Recognizer Output Voting Error Reduction (ROVER) (Fiscus, 1997) ensemble strategy combining outputs from three distinct SOTA models as shown in Figure 1(g). We align transcripts at the word level and apply a prioritized majority voting scheme: a word is accepted if predicted by at least two models; otherwise, we default to the prediction of our primary backbone, Whisper, to maintain consistency. Residual hallucinations are further pruned using a RepetitionFilter that discards samples with excessive n-gram ( $n = 15$ ) repetitions (count  $\geq 5$ ) (Udandarao et al., 2025). Concurrently, we extract word-level timestamps via Whisper. Precision in temporal alignment is critical for modern streaming speech language models (as detailed in Section 3.1), which typically require strict synchronization between audio and text tokens.

# 3 Experiments

In this section, we validate the individual components of our proposed pipeline. First, we examine the practical utility of our approach by fine-tuning Moshi using the dataset preprocessed by our proposed pipeline (§ 3.1). We then quantitatively evaluate the diarization accuracy (§ 3.2), the audio quality following overlap separation (§ 3.3), and the accuracy of the ensemble-based ASR (§ 3.4). Furthermore, to provide a comprehensive analysis, we discuss the pipeline’s latency (§ 3.5).

## 3.1 Effectiveness of Sommelier-Processed Data for Full-Duplex Models

To validate the effectiveness of our proposed **Sommelier** pipeline, we examine whether training a full-duplex model on the data processed by this pipeline yields performance improvements. To this end, we performed LoRA fine-tuning on

| Model         | DER (%)     | JER (%)      | DER<br>( $\leq 1.0s$ , %) | DER<br>(turn, %) |
|---------------|-------------|--------------|---------------------------|------------------|
| pyannote3.1   | 8.40        | 17.68        | 20.21                     | 0.051            |
| sortformer_v1 | <b>7.16</b> | <b>14.69</b> | <b>16.87</b>              | <b>0.006</b>     |

**Table 2** Diarization model ablation on VoxConverse (Chung et al., 2020) (common subset,  $\leq 4$  speakers). Lower is better for DER/JER and RTF.

| SIR   | OVL | WER (%) ↓ |             |     | STOI ↑ |             |      | UTMOS ↑ |             |      |
|-------|-----|-----------|-------------|-----|--------|-------------|------|---------|-------------|------|
|       |     | Ori       | Sep         | Orc | Ori    | Sep         | Orc  | Ori     | Sep         | Orc  |
| 0 dB  | 0.2 | 10.5      | <b>6.1</b>  | 4.8 | .961   | <b>.982</b> | 1.00 | 3.04    | <b>3.53</b> | 3.88 |
|       | 0.5 | 13.9      | <b>7.9</b>  | 5.8 | .888   | <b>.969</b> | 1.00 | 2.27    | <b>3.32</b> | 3.87 |
|       | 1.0 | 48.9      | <b>15.6</b> | 5.3 | .778   | <b>.913</b> | 1.00 | 1.70    | <b>3.02</b> | 3.84 |
| 5 dB  | 0.2 | 11.3      | <b>7.6</b>  | 5.3 | .961   | <b>.978</b> | 1.00 | 3.06    | <b>3.47</b> | 3.87 |
|       | 0.5 | 18.8      | <b>7.1</b>  | 4.3 | .887   | <b>.971</b> | 1.00 | 2.34    | <b>3.39</b> | 3.91 |
|       | 1.0 | 52.5      | <b>9.1</b>  | 4.0 | .761   | <b>.936</b> | 1.00 | 1.79    | <b>3.12</b> | 3.91 |
| 10 dB | 0.2 | 12.6      | <b>7.0</b>  | 5.6 | .961   | <b>.980</b> | 1.00 | 3.26    | <b>3.60</b> | 3.98 |
|       | 0.5 | 29.7      | <b>10.1</b> | 5.2 | .877   | <b>.956</b> | 1.00 | 2.58    | <b>3.21</b> | 3.86 |
|       | 1.0 | 51.0      | <b>13.8</b> | 4.8 | .754   | <b>.919</b> | 1.00 | 2.17    | <b>3.01</b> | 3.92 |

**Table 3** Speech quality for separated overlapped speech across metrics for **Original** audio, source **Separated**, and **Oracle** (pre-synthesis speech quality).

moshiko-pytorch-bf16 (Défossez et al., 2024) and evaluated its duplex performance using the Full-Duplex-Bench (Lin et al., 2025b,a).

**Dataset.** We find that long turn-taking in the training data for Moshi, specifically when a single speaker holds the floor for too long (more than a minute), leads to unstable loss reduction and degrades performance, causing the model to become unresponsive. Consequently, we selected segments from the **Sommelier**-processed data where each turn lasted no more than 10 seconds. We defined a valid region as a sequence of at least three consecutive turns, truncating the region if an utterance exceeding 10 seconds appeared. Additionally, we assigned only a single speaker to the left channel of the stereo training data. We confirmed that these selection criteria significantly impact the training dynamics. The model configuration is detailed in Appendix D.

**Results.** As shown in Table 1, evaluation on Full-Duplex-Bench 1.0 demonstrated performance improvements across Backchanneling, Smooth Turn-Taking, and User Interruption handling. Regarding Pause Handling, however, we observe that the model performs comparably to base Moshi, exhibiting similar limitations. We hypothesize that this stems from the Moshi architecture or the absence of prompt audio, as proposed in Personaplex (Roy et al., 2026). Regarding latency, the base model exhibited notably short latencies simply because it failed to engage in backchanneling or interruption handling, reflecting suboptimal behavior where the model continued speaking regardless of user input. In contrast, after fine-tuning with Sommelier-processed data, the increased latency can be interpreted positively, as it indicates that the model is now actively processing user input and preparing appropriate responses for backchannels and interruptions. Detailed descriptions of the benchmark metrics are provided in Appendix D.

### 3.2 Diarization Model Choice

Using Pyannote 3.1 has been widely regarded as the default for diarization models, a trend followed by recent works such as He et al. (2024). In this study, however, we compare the performance of Sortformer (Park et al.), which is adopted in our pipeline, against the Pyannote 3.1 (Bredin, 2023; Plaquet and Bredin, 2023) baseline.

**Metrics.** We evaluate speaker diarization quality using DER (Diarization Error Rate) and JER (Jaccard Error Rate). DER measures the fraction of speaker time that is incorrectly attributed, typically aggregating *missed speech*, *false alarm speech*, and *speaker confusion* within a tolerance collar. JER measures the average Jaccard distance between the reference and hypothesis speaker segments, and is known to be more sensitive to boundary quality and segmentation consistency. To stress-test challenging regimes, we additionally compute *DER on short-duration speech* by restricting evaluation to reference segments shorter than a threshold ( $\leq 0.5s$  and  $\leq 1.0s$ ), and *DER on turn-taking regions* by

| Dataset                   | Model             | WER (%)           | Time (s) |
|---------------------------|-------------------|-------------------|----------|
| LibriSpeech<br>Test Clean | Whisper           | $3.63 \pm 9.37$   | 0.39     |
|                           | <b>MoE (Ours)</b> | $2.04 \pm 6.50$   | 1.40     |
| LibriSpeech<br>Test Other | Whisper           | $6.26 \pm 11.63$  | 0.35     |
|                           | <b>MoE (Ours)</b> | $3.92 \pm 8.92$   | 1.27     |
| TEDLIUM3<br>Test          | Whisper           | $12.19 \pm 12.31$ | 0.36     |
|                           | <b>MoE (Ours)</b> | $10.66 \pm 11.73$ | 1.33     |

**Table 4** Evaluation results on LibriSpeech (Clean/Other) and TEDLIUM3. Whisper refers Whisper-large-v3 model.

restricting evaluation to temporal windows around speaker change points (speaker alternations within a small gap). All metrics are reported on the VoxConverse (Chung et al., 2020) common subset containing recordings with at most four speakers.

**Results and analysis.** Table 2 demonstrates that Sortformer consistently outperforms the Pyannote 3.1 baseline across global metrics on the VoxConverse benchmark. More importantly, the gains are most pronounced in regimes critical for conversational modeling. Sortformer exhibits superior robustness in handling short utterances and rapid turn-taking, effectively reducing errors in brief interjections and speaker boundaries. These results confirm that Sortformer is better suited for processing highly interactive, overlapping dialogue than standard baselines.

### 3.3 Speech Quality of Overlap Separation

Processing overlapped speech is a critical step in constructing training datasets for full-duplex conversational models (Défossez et al., 2024). This is because full-duplex training requires speech segments to overlap freely, as in natural human conversations, while maintaining source-separated audio streams for each speaker.

Given two speech segments  $a_i$  and  $a_j$  that are sequentially overlapped, where  $a_i$  starts at  $t_{start}$ ,  $a_j$  ends at  $t_{end}$ , and the overlap occurs from  $t_1$  to  $t_2$ , we evaluate speech quality for each diarized speaker’s utterance interval:  $[t_{start}, t_2]$  for speaker1 and  $[t_1, t_{end}]$  for speaker2.

**Dataset.** To simulate diverse real-world overlap conditions, we synthesized 900 samples of two speaker mixtures from the LibriSpeech (Panayotov et al., 2015) test utterance by varying Signal-to-Interference Ratio ( $SIR \in \{0, 5, 10\}$  dB) and overlap ratio ( $\rho \in \{0.2, 0.5, 1.0\}$ ), forming nine different conditions. We also mix silence-trimmed sources to achieve the target overlap precisely.

**Metrics.** Evaluation is conducted across three conditions: (1) *Original*, which directly extracts time segments from the mixed signal, (2) **Sep**, which applies SepReformer (Shin et al., 2024)-based separation with speaker identity matching (see Section 2.3), and (3) *Oracle*, which uses clean source signals as an upper bound. Ground-truth diarization timestamps from data synthesis are used across all conditions to ensure fair evaluation of overlapped regions. We assess intelligibility using Word Error Rate (WER), acoustic quality using SI-SDR and STOI, and perceptual naturalness using UTMOS (Saeki et al., 2022).

**Results and Analysis.** The quantitative analysis presented in Table 3 reveals that while variations in the Signal-to-Interference Ratio (SIR) have a limited impact on performance, the overlap ratio serves as the critical determinant of task difficulty. As the overlap ratio increases, the baseline model suffers significant degradation; however, our proposed method (Sep) consistently outperforms the baseline across all experimental conditions, demonstrating robust separation capabilities even in highly overlapped scenarios. Most notably, in terms of perceptual quality (UTMOS (Saeki et al., 2022)), the proposed method achieves scores closely approximating the Oracle upper bound. This result strongly suggests that our model not only improves intelligibility but also preserves speech naturalness effectively, thereby guaranteeing the generation of high-quality samples suitable for use as training data. More detailed results and analysis are provided in Appendix B.



| Stage                  | Processing Time (s) | RTF           |
|------------------------|---------------------|---------------|
| Audio Duration         | 120.00              | –             |
| VAD + Sortformer       | 1.91                | 0.0159        |
| SepReformer Separation | 0.15                | 0.0013        |
| ASR ensemble           | 13.91               | 0.1159        |
| FlowSE Denoising       | 4.99                | 0.0416        |
| <b>Total</b>           | <b>20.95</b>        | <b>0.1746</b> |

**Table 5** Processing time breakdown for the proposed pipeline on a 120-second audio sample.

### 3.4 ASR Ensemble Performance

We compare the performance of the Whisper model against our proposed three-model ensemble method utilizing ROVER [Fiscus \(1997\)](#).

**Metrics.** To evaluate ASR performance, we measure the Word Error Rate (WER) using the LibriSpeech ([Panayotov et al., 2015](#)) test dataset (clean and other splits) and the TEDLIUM3 ([Hernandez et al., 2018](#)) test set. Since LibriSpeech test-other and TEDLIUM3 contain noisy conditions, these datasets allow us to assess the model’s robustness against real-world scenarios.

**Results and Analysis.** The comparison results between the standalone Whisper Large v3 and the ensemble (Whisper + Canary + Parakeet ([Sekoyan et al., 2025](#))) are presented in Table 4. In terms of WER, the proposed approach demonstrated a significant improvement of approximately 37%, reducing the error rate from 6.26% to 3.92% compared to the single Whisper-large-v3 baseline. This gap was particularly evident in noisy data, demonstrating improved recognition accuracy in segments containing low volume or BGM.

Regarding inference time, the ensemble approach required approximately three times longer than the baseline. This latency is primarily attributed to the inference speed of Canary, the slowest model among the three, rather than sequential execution. Additionally, the concurrent loading and inference of three models may introduce slight overhead.

Qualitatively, we observed hallucinations in the Whisper outputs, such as repetitive generation (e.g., “*Yeah., Yeah., Yeah...*”). We confirmed that our method successfully corrected these errors by selecting the accurate transcript provided by Canary (e.g., “*Yeah, big decision for Dan*”).

### 3.5 Latency

Data preprocessing is a computationally intensive task [He et al. \(2024\)](#); [Dua et al. \(2025\)](#). Thus, minimizing latency in this phase is critical. As shown in Table 5, running a single process on an A100 (80GB) yields a total Real-Time Factor (RTF) of 0.1746. Excluding the optional FlowSE Denoising step further reduces the RTF to 0.133, with the primary bottleneck occurring in the ASR stage. Given the peak memory usage of 23GB, it is possible to allocate three concurrent processes on a single GPU, which effectively lowers the RTF to 0.0443 per GPU. Consequently, processing 10,000 hours of audio using eight A100 GPUs would take approximately 55 hours, demonstrating the practical feasibility of our approach.

## 4 Conclusion

We presented **Sommelier**, the first scalable, open-source pipeline for full-duplex SLMs. Our pipeline combines rigorous diarization, overlap handling, and ensemble-based ASR to improve overall transcript quality. We validated the overall utility of the **Sommelier** pipeline by fine-tuning Moshi on **Sommelier**-processed speech. We release our pipeline to support reproducible industrial research and to accelerate progress toward natural, real-time human–AI interaction.

### Limitations

A limitation of our pipeline is its exclusive focus on processing speech data. While optimized for conversational dialogue, it does not explicitly account for non-speech acoustic events or general sound scenes, limiting its scope compared to

omni-modal audio approaches. Although our overlap separation module effectively disentangles simultaneous speakers from single-stream recordings, the resulting audio fidelity is inevitably slightly inferior to datasets that are originally recorded with distinct, isolated channels (Oracle), as the artificial separation process may introduce minor acoustic artifacts.

## Ethical Considerations

We developed the **Sommelier** pipeline with a strict adherence to open-source compliance and intellectual property rights. All software components, libraries, and pre-trained models integrated into our framework are governed by commercially permissive licenses, primarily MIT and Creative Commons (CC), allowing for broad academic and industrial application without legal ambiguity.

Furthermore, the podcast audio samples featured on our project demonstration page were exclusively selected from sources explicitly released under CC licenses. We have rigorously verified the usage terms of these recordings to ensure that no copyrighted material is infringed upon and that the original creators' rights are respected.

Beyond licensing compliance, we acknowledge the broader implications of releasing tools for high-fidelity speech processing. While our goal is to advance full-duplex interaction, we recognize that high-quality conversational datasets can potentially be misused for non-consensual voice cloning or deepfake generation. We urge the research community to utilize this pipeline responsibly, ensuring that any private data processed is done so with appropriate consent and privacy safeguards in place.

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## Appendix

We provide detailed supplementary materials organized as follows:

- **Appendix A** reviews related work on full-duplex models, large-scale speech datasets, and speech preprocessing techniques.
  - **Appendix B** presents additional experimental results regarding overlap separation.
  - **Appendix C** illustrates specific cases for handling backchanneling and overlapping speech.
  - **Appendix D** details supplementary results from the fine-tuning experiments.
  - **Appendix E** describes the techniques employed for audio captioning.
  - **Appendix F** provides examples of data processed by **Sommelier**.
- 

## A Related Works

In this section, we provide a comprehensive overview of the research landscape relevant to our work. We begin by tracing the evolution of Speech Language Models from cascaded pipelines to end-to-end architectures, with a particular focus on the recent shift toward full-duplex interaction (§A.1). We then critically examine the current landscape of large-scale speech datasets, identifying their structural limitations for modeling naturalistic conversational dynamics (§A.2). Finally, we survey the automated pipelines used for speech data curation, discussing key technical challenges such as speaker diarization errors and transcription hallucinations that our work aims to address (§A.3).

### A.1 Speech Language Models and Full-Duplex Interaction

The landscape of spoken language understanding has undergone a fundamental transformation, moving away from cascaded systems towards end-to-end modeling. Traditional conversational agents relied on a cascade pipeline of Automatic Speech Recognition (ASR), Large Language Models (LLM), and Text-to-Speech (TTS) (Lee et al., 2025; Jung et al., 2024a). While effective for distinct tasks, this modular approach inevitably suffers from error propagation and the loss of paralinguistic features such as emotion, prosody, and intonation. To address these limitations, End-to-End Speech Language Models (SLMs), such as AudioLM (Borsos et al., 2023) and AudioPaLM (Rubenstein et al., 2023), were introduced to process acoustic tokens directly, preserving the rich nuances of speech. Building on this foundation, recent multimodal models like Qwen-Audio Chu et al. (2023, 2024), Qwen-Omni Xu et al. (2025a,b), HyerCLOVA-X-8B-Omni (Team, 2026b,a) and Audio Flamingo Chu et al. (2024); Goel et al. (2025) have demonstrated exceptional capabilities in understanding and reasoning across both text and audio modalities, effectively bridging the gap between sound processing and language comprehension.

Despite these advancements, a critical frontier remains in achieving natural, real-time human-computer interaction. While earlier models primarily operated on a turn-based (half-duplex) mechanism, recent developments like Moshi Défossez et al. (2024) and GPT-4o Hurst et al. (2024) aim to realize full-duplex communication, where listening and speaking occur simultaneously. This shift necessitates models to master complex conversational dynamics, including the ability to handle overlapping speech, process back-channeling (e.g., rapid affirmations), and predict turn-taking opportunities seamlessly. However, training such full-duplex systems presents a significant challenge: it requires high-quality, multi-stream data that captures these intricate acoustic intersections. The scarcity of such datasets in the current research landscape creates a bottleneck, limiting the ability of current models to generalize to the chaotic and fluid nature of real-world dialogue.

### A.2 Large-Scale Speech Datasets

Existing speech datasets, despite their increasing volume, remain suboptimal for training full-duplex models that require rich interactional dynamics. Traditional ASR benchmarks like LibriSpeech (Panayotov et al., 2015) and GigaSpeech (Chen et al., 2021) are dominated by scripted read speech or solitary monologues, failing to capture the

dynamic and interactive spontaneity of human dialogue. While conversational datasets such as Fisher (Cieri et al., 2004) and Switchboard (Godfrey and Holliman, 1993) offer multi-speaker interactions, they are severely limited by their archaic telephony quality (8kHz), narrow bandwidth, and relatively small scale (typically a few thousand hours), which is insufficient for modern large-scale pre-training Radford et al. (2022b); Xu et al. (2025b). While recent web-scale initiatives like WenetSpeech (Zhang et al., 2022) and Emilia He et al. (2024) have successfully aggregated massive datasets, their pipelines are heavily optimized for single-stream speech, thereby neglecting the concurrent dynamics required for full-duplex interaction. Crucially, their pre-processing pipelines treat overlapping speech as noise to be excised or ignored rather than a feature to be modeled. This structural limitation results in data that lacks the distinct multi-stream separation and essential acoustic collisions required for learning true full-duplex interaction.

### A.3 Automated Speech Data Processing Pipelines

While open-source data processing pipelines have become the bedrock of Large Language Model (LLM) research, exemplified by transparent frameworks like Dolma (Soldaini et al., 2024), RedPajama (Weber et al., 2024), and FineWeb (Penedo et al., 2024), the domain of speech processing remains significantly opaque. Although model weights for Speech Language Models (SLMs) are frequently released, the intricate “data recipes” required to curate high-quality pre-training corpora remain proprietary “black boxes,” impeding the community’s ability to reproduce results or improve upon existing strategies.

This lack of standardized, open pipelines is particularly critical when addressing the technical demands of full-duplex communication. Current methodologies rely heavily on tools designed for single-stream processing (Dua et al., 2025; Yang et al., 2025), which are ill-suited for capturing the concurrent dynamics of human dialogue. For instance, while speaker diarization is a prerequisite for multi-turn modeling, standard tools like Pyannote (Bredin, 2023; Plaquet and Bredin, 2023) often struggle in the complex acoustic environments of in-the-wild web videos. Crucially, these tools frequently misinterpret the *overlaps* and rapid *turn-taking*, essential features of full-duplex interaction, as segmentation errors or noise, thereby degrading the structural integrity of the conversational data.

Furthermore, the reliance on ASR models for transcription introduces the risk of hallucinations. Models like Whisper (Radford et al., 2022b), though powerful, are prone to generating repetitive loops or nonsensical text during silence or non-speech intervals, a critical instability highlighted in recent studies such as *Careless Whisper* (Koencke et al., 2024a). Existing pipelines lack the robustness to filter these hallucinations or handle the multi-stream nature of duplex speech, underscoring the urgent need for a transparent, hallucination-aware processing framework tailored for conversational AI.

## B Detail of Overlapping disentangle experiments

This section presents detailed experimental results on the efficacy of our overlap separation module. We evaluated performance across varying Signal-to-Interference Ratios (SIR) and Overlap Ratios using four key metrics: Word Error Rate (WER), Scale-Invariant Signal-to-Distortion Ratio (SI-SDR), Short-Time Objective Intelligibility (STOI), and UTMOS. The results, summarized in Tables 6 through 8, demonstrate that applying the separation module (‘Sep’) consistently improves signal quality compared to the baseline (‘Base’). Notably, the performance gain is significantly larger for Speaker 2 (the interfering or secondary speaker) than for Speaker 1 (the primary speaker), particularly in challenging conditions with high overlap ratios as shown in Figure 3.

### B.1 Analysis of Results

*Asymmetric Gains (Spk1 vs. Spk2)* Across all metrics, the gap between the ‘Base’ and ‘Sep’ conditions is most dramatic for Speaker 2. For instance, in the 0,dB SIR and 1.0 overlap condition, the WER for Speaker 2 improves drastically from 0.444 to 0.138 (Table 6), whereas Speaker 1 sees a relatively smaller, though still significant, improvement. This suggests that our module is particularly effective at recovering the subordinate or quieter speaker in a mixture, which is crucial for full-duplex conversational AI where both parties must be heard clearly.

*Resilience to High Overlap* The benefits of separation become more pronounced as the overlap ratio increases. In the worst-case scenario (1.0 overlap), the baseline UTMOS scores (Table 8) drop severely (e.g.,  $\approx 1.7$ ), but the separation module restores quality to near-natural levels ( $\approx 3.0$ ). Similarly, STOI scores (Table 7) remain high ( $> 0.9$ ) even under full overlap when separation is applied, confirming that intelligibility is preserved.

| SIR   | Overlap | Speaker 1    |              |        | Speaker 2 |              |        |
|-------|---------|--------------|--------------|--------|-----------|--------------|--------|
|       |         | Base         | Sep          | Oracle | Base      | Sep          | Oracle |
| 0 dB  | 0.2     | 0.109        | <b>0.048</b> | 0.034  | 0.100     | <b>0.074</b> | 0.061  |
|       | 0.5     | 0.162        | <b>0.094</b> | 0.078  | 0.115     | <b>0.065</b> | 0.039  |
|       | 1.0     | 0.535        | <b>0.175</b> | 0.058  | 0.444     | <b>0.138</b> | 0.048  |
| 5 dB  | 0.2     | <b>0.080</b> | 0.088        | 0.066  | 0.146     | <b>0.065</b> | 0.040  |
|       | 0.5     | 0.099        | <b>0.059</b> | 0.036  | 0.277     | <b>0.084</b> | 0.049  |
|       | 1.0     | 0.136        | <b>0.069</b> | 0.039  | 0.913     | <b>0.113</b> | 0.042  |
| 10 dB | 0.2     | 0.059        | <b>0.056</b> | 0.058  | 0.194     | <b>0.084</b> | 0.053  |
|       | 0.5     | 0.067        | <b>0.064</b> | 0.051  | 0.527     | <b>0.138</b> | 0.052  |
|       | 1.0     | 0.096        | <b>0.083</b> | 0.051  | 0.923     | <b>0.193</b> | 0.044  |

**Table 6** Word Error Rate (WER) comparison across different conditions. Lower is better.

| SIR   | Overlap | Speaker 1 |              |        | Speaker 2 |              |        |
|-------|---------|-----------|--------------|--------|-----------|--------------|--------|
|       |         | Base      | Sep          | Oracle | Base      | Sep          | Oracle |
| 0 dB  | 0.2     | 0.959     | <b>0.980</b> | 1.000  | 0.964     | <b>0.985</b> | 1.000  |
|       | 0.5     | 0.889     | <b>0.969</b> | 1.000  | 0.887     | <b>0.968</b> | 1.000  |
|       | 1.0     | 0.785     | <b>0.918</b> | 1.000  | 0.771     | <b>0.908</b> | 1.000  |
| 5 dB  | 0.2     | 0.971     | <b>0.980</b> | 1.000  | 0.951     | <b>0.976</b> | 1.000  |
|       | 0.5     | 0.929     | <b>0.980</b> | 1.000  | 0.844     | <b>0.962</b> | 1.000  |
|       | 1.0     | 0.847     | <b>0.955</b> | 1.000  | 0.676     | <b>0.917</b> | 1.000  |
| 10 dB | 0.2     | 0.985     | <b>0.988</b> | 1.000  | 0.938     | <b>0.971</b> | 1.000  |
|       | 0.5     | 0.956     | <b>0.981</b> | 1.000  | 0.798     | <b>0.931</b> | 1.000  |
|       | 1.0     | 0.901     | <b>0.954</b> | 1.000  | 0.608     | <b>0.883</b> | 1.000  |

**Table 7** Short-Time Objective Intelligibility (STOI) scores. Higher is better.

the Baseline method exhibits competitive SI-SDR performance at  $\rho \in \{0.2, 0.5\}$ . However, this comparison is inherently unfair: the Baseline extracts each speaker’s time segment directly from the mixed signal, meaning that at 20% and 50% overlap ratios, the majority of each segment contains clean, non-overlapping speech. For instance, at  $\rho = 0.2$ , the actual overlap constitutes only 15.2% for S1 and 14.7% for S2, leaving approximately 7 seconds of clean audio per speaker. Consequently, ASR systems can effectively recognize the clean portions, resulting in artificially low WER for the Baseline.

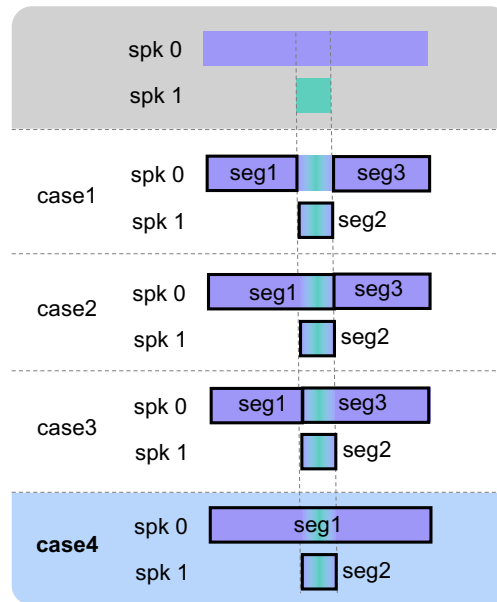
## C Overlap cases

In this section, we present two representative overlap cases. The first is *backchanneling*, where a speaker produces a short utterance while the other is speaking; in this case, one segment is fully contained within another segment. The second is *overlap*, where two segments partially overlap but neither segment fully contains the other.

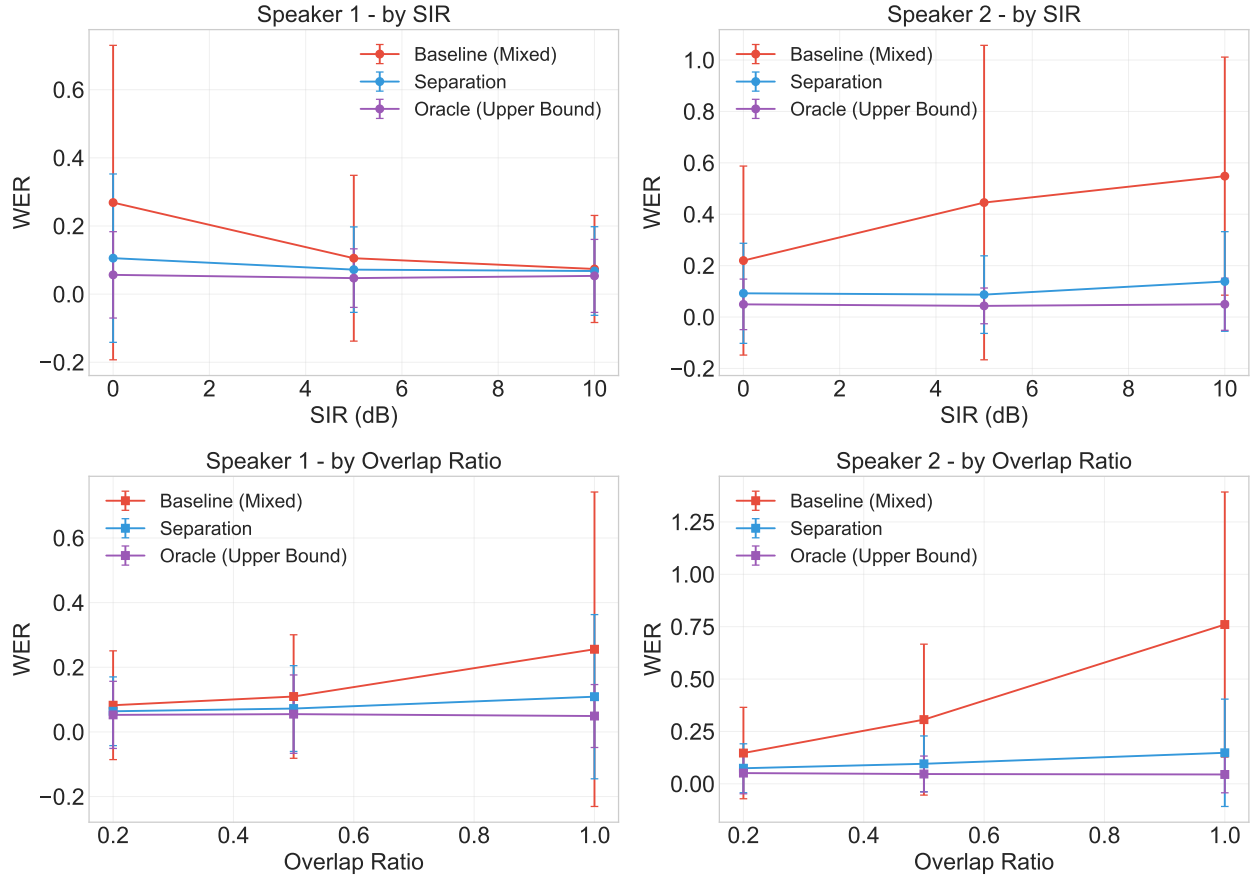


| SIR   | Overlap | Speaker 1 |             |        | Speaker 2 |             |        |
|-------|---------|-----------|-------------|--------|-----------|-------------|--------|
|       |         | Base      | Sep         | Oracle | Base      | Sep         | Oracle |
| 0 dB  | 0.2     | 3.08      | <b>3.56</b> | 3.94   | 2.99      | <b>3.50</b> | 3.82   |
|       | 0.5     | 2.28      | <b>3.29</b> | 3.85   | 2.25      | <b>3.35</b> | 3.90   |
|       | 1.0     | 1.73      | <b>3.05</b> | 3.81   | 1.67      | <b>2.99</b> | 3.86   |
| 5 dB  | 0.2     | 3.12      | <b>3.47</b> | 3.83   | 3.00      | <b>3.48</b> | 3.90   |
|       | 0.5     | 2.46      | <b>3.51</b> | 3.90   | 2.22      | <b>3.28</b> | 3.92   |
|       | 1.0     | 1.86      | <b>3.29</b> | 3.94   | 1.72      | <b>2.94</b> | 3.87   |
| 10 dB | 0.2     | 3.39      | <b>3.69</b> | 3.95   | 3.12      | <b>3.51</b> | 4.02   |
|       | 0.5     | 2.82      | <b>3.50</b> | 3.91   | 2.35      | <b>2.92</b> | 3.82   |
|       | 1.0     | 2.22      | <b>3.29</b> | 3.95   | 2.12      | <b>2.73</b> | 3.90   |

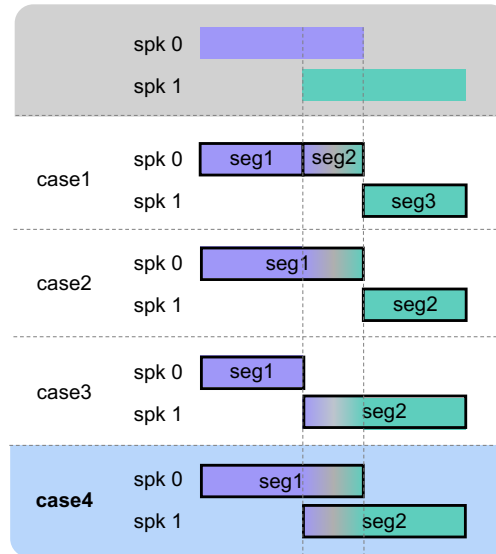
**Table 8** UTMOS (MOS prediction) scores. Higher is better.



**Figure 4** Four ways to handle backchanneling in overlapping speech.



**Figure 3** WER comparison by method, SIR, and overlap ratio for both speakers. Top: WER as a function of SIR (dB). Bottom: WER as a function of overlap ratio. Methods include Baseline (mixed), Separation, and Oracle. Error bars represent standard deviation.



**Figure 5** Four distinct types of separable cases in overlapping speech.

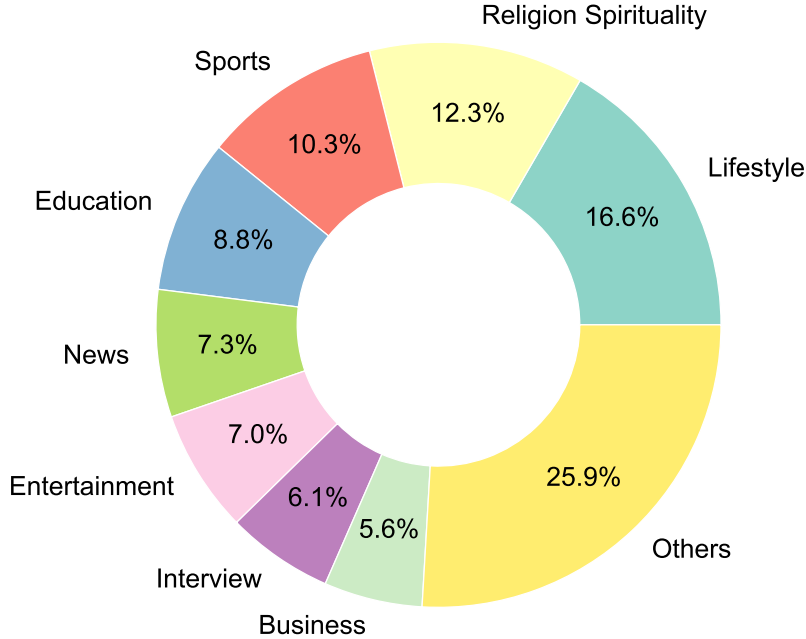


Figure 6 Category-wise statistics of the dataset used for Moshi fine-tuning experiments.

## D Detail of Finetuning Experiment

The fine-tuning hyperparameters for Moshiko are listed in Table 9. We significantly benefited from the implementation provided at <https://github.com/kyutai-labs/moshi-finetune>.

| Hyperparameter      | Value                  |
|---------------------|------------------------|
| Total Data Duration | $\approx 83$ hours     |
| Training Steps      | 2,000                  |
| Hardware            | $8 \times \text{A100}$ |
| Rank                | 128                    |
| Batch Size          | 16                     |
| Learning Rate       | $2e-6$                 |
| Weight Decay        | 0.1                    |

Table 9 Training hyperparameters and settings.

### D.1 Dataset Statistics

This section provides statistics for the data fine-tuned in Section 3.1. Figure 6 illustrates that our training data originates from a wide range of conversational domains.

### D.2 Full-Duplex-Bench 1.0: Metric Definitions

Full-Duplex-Bench 1.0 evaluates spoken dialogue models under full-duplex conditions, focusing on pause handling, backchanneling, smooth turn-taking, and user interruption handling. Across all tasks, we define latency for an instance  $i$  as  $\Delta_i = t_{\text{start},i} - t_{\text{end},i}$ , where  $t_{\text{start}}$  denotes the model’s response onset and  $t_{\text{end}}$  denotes the end of the relevant user event.

**Table 10** Full-Duplex-Bench v1.5 Evaluation Results (Moshi vs. Fine-tuned Moshi). Comparison of post-distractor audio quality metrics and behavior classification across four overlap scenarios.

| Full-Duplex-Bench v1.5: Overlap Handling Evaluation |                      |                 |                   |                  |       |       |          |           |          |                |      |        |      |
|-----------------------------------------------------|----------------------|-----------------|-------------------|------------------|-------|-------|----------|-----------|----------|----------------|------|--------|------|
| Scenario                                            | Audio Quality (Post) |                 |                   |                  | Rate  | Pitch |          | Intensity |          | Behavior Ratio |      |        |      |
|                                                     | STOI $\uparrow$      | PESQ $\uparrow$ | SI-SDR $\uparrow$ | UTMOS $\uparrow$ | WPM   | $\mu$ | $\sigma$ | $\mu$     | $\sigma$ | RESP           | RESU | UNCERT | UNK  |
| <i>Moshi (Base)</i>                                 |                      |                 |                   |                  |       |       |          |           |          |                |      |        |      |
| Background Speech                                   | 0.79                 | 2.19            | 5.43              | 1.86             | 75.9  | 85.9  | 11.5     | -64.6     | 16.3     | 0.15           | 0.07 | 0.03   | 0.75 |
| Talking to Other                                    | 0.90                 | 2.55            | 12.64             | <b>2.34</b>      | 124.5 | 96.6  | 16.0     | -49.1     | 16.2     | 0.15           | 0.18 | 0.04   | 0.63 |
| User Backchannel                                    | 0.63                 | 1.60            | -6.57             | 1.25             | 25.8  | 66.5  | 6.2      | -87.0     | 16.0     | 0.01           | 0.06 | 0.01   | 0.92 |
| User Interruption                                   | 0.94                 | 2.87            | 16.07             | <b>2.65</b>      | 149.0 | 111.1 | 21.8     | -41.3     | 16.0     | 0.59           | 0.17 | 0.03   | 0.21 |
| <i>Moshi (Fine-tuned)</i>                           |                      |                 |                   |                  |       |       |          |           |          |                |      |        |      |
| Background Speech                                   | <b>0.98</b>          | <b>3.33</b>     | <b>20.76</b>      | <b>1.87</b>      | 157.5 | 88.3  | 13.7     | -64.8     | 16.4     | 0.28           | 0.11 | 0.00   | 0.60 |
| Talking to Other                                    | <b>0.96</b>          | <b>3.30</b>     | <b>20.26</b>      | 2.30             | 146.4 | 96.4  | 17.0     | -51.0     | 16.3     | 0.18           | 0.16 | 0.00   | 0.63 |
| User Backchannel                                    | <b>0.91</b>          | <b>3.01</b>     | <b>16.48</b>      | <b>1.32</b>      | 132.5 | 72.0  | 7.9      | -85.2     | 16.0     | 0.08           | 0.11 | 0.00   | 0.72 |
| User Interruption                                   | <b>0.97</b>          | <b>3.27</b>     | <b>20.26</b>      | 2.58             | 156.0 | 110.7 | 22.8     | -43.8     | 15.7     | 0.51           | 0.12 | 0.00   | 0.36 |

*Pause Handling.* To evaluate whether the model incorrectly treats mid-utterance silence as a turn boundary, we use **Synthetic/Candor TOR** (Turn-Over Rate,  $\downarrow$ ). This metric calculates the fraction of pause instances in which the model starts speaking. A failure is recorded if the model’s output during a pause exceeds a minimal threshold (duration  $\geq 1$  second or  $> 3$  words).

*Backchanneling.* We assess the model’s ability to provide brief acknowledgements without seizing the floor using three metrics. **Backchannel TOR** ( $\downarrow$ ) measures the fraction of backchannel-eligible windows where the model produces a full turn (duration  $\geq 3$ s or  $\geq 1$ s with  $> 3$  words). **Frequency** ( $\uparrow$ ) reports the number of backchannels normalized by total audio duration. Finally, **JSD** ( $\downarrow$ ) computes the Jensen–Shannon divergence between the model’s backchannel timing distribution and human ground truth to evaluate timing naturalness.

*Smooth Turn Taking.* This task measures the model’s promptness in responding after the user completes an utterance. We report **Candor TOR** ( $\uparrow$ ), defined as the fraction of user-turn endings where the model successfully begins speaking, and **Latency** ( $\downarrow$ ), measured only on instances where the model successfully takes the turn.

*User Interruption.* When a user interrupts the model, we evaluate the system’s responsiveness and contextual adaptation. **Interruption TOR** ( $\uparrow$ ) measures the fraction of interruption events where the model responds. **Latency** ( $\downarrow$ ) tracks the time from the end of the interruption to the model’s response. Additionally, we use a **GPT-4o relevance score** ( $\uparrow$ ) (0–5 scale) to assess whether the model’s response is semantically relevant to the content of the interruption.

### D.3 Results on Full-Duplex-Bench 1.5

Lin et al. (2025a) released Full-Duplex-Bench 1.5 as a successor to v1.0. We further evaluated the model fine-tuned on **Sommelier**-processed data using this updated benchmark.

The experimental results presented in Table 10 and Table 11 demonstrate that the **Sommelier**-fine-tuned Moshi model significantly outperforms the base model across all overlap scenarios. In terms of audio quality, the fine-tuned model exhibits superior signal fidelity and robustness, evidenced by substantial gains in PESQ and SI-SDR scores; notably, the SI-SDR for the ‘Background Speech’ scenario improved drastically from 5.43 dB to 20.76 dB. Furthermore, the latency analysis reveals a critical enhancement in conversational responsiveness, with both stop and response latencies reduced to sub-second averages in the majority of cases, thereby enabling more natural and immediate turn-taking interactions.

## E Context Captioning

Speech data contains rich non-verbal information, such as timbre and emotion, beyond text semantics (Koolagudi and Rao, 2012; Tursunov et al., 2019; Lee et al., 2025; Jung et al., 2024b). Detailed captioning of this information serves as effective metadata for speech understanding and generation (Wang et al., 2025a; AI et al., 2025; Yan et al., 2025).

**Table 11** Full-Duplex-Bench v1.5 Latency Analysis (Moshi vs. Fine-tuned Moshi). Stop latency measures time from user speech onset to model speech cessation. Response latency measures time from user speech offset to model speech resumption.

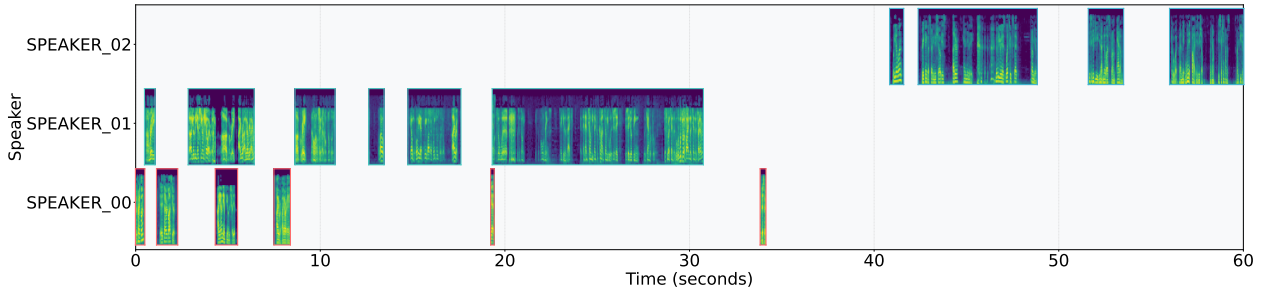
| Latency Analysis (seconds) |               |             |                   |             |              |      |
|----------------------------|---------------|-------------|-------------------|-------------|--------------|------|
| Scenario                   | Stop Latency↓ |             | Response Latency↓ |             | Sample Count |      |
|                            | $\mu$         | $\sigma$    | $\mu$             | $\sigma$    | Stop         | Resp |
| <i>Moshi (Base)</i>        |               |             |                   |             |              |      |
| Background Speech          | 1.02          | 0.55        | 2.90              | 2.05        | 150          | 89   |
| Talking to Other           | 1.13          | <b>0.57</b> | 3.22              | 1.87        | 184          | 117  |
| User Backchannel           | 1.30          | 0.42        | 2.38              | 1.84        | 113          | 29   |
| User Interruption          | 1.30          | 0.72        | 1.99              | 2.24        | 391          | 237  |
| <i>Moshi (Fine-tuned)</i>  |               |             |                   |             |              |      |
| Background Speech          | <b>0.68</b>   | <b>0.48</b> | <b>0.73</b>       | <b>0.49</b> | 44           | 192  |
| Talking to Other           | <b>0.82</b>   | 0.65        | <b>0.84</b>       | <b>0.69</b> | 47           | 188  |
| User Backchannel           | <b>0.70</b>   | <b>0.40</b> | <b>1.12</b>       | <b>0.85</b> | 57           | 156  |
| User Interruption          | <b>0.89</b>   | <b>0.64</b> | <b>0.66</b>       | <b>0.55</b> | 110          | 383  |

Unlike other studies, we propose captioning audio segments using the Qwen3-Omni-Captioner (Xu et al., 2025b) model to generate rich metadata, including emotion, gender, age group, and situation descriptions.

However, captioning short segments individually can fail to capture context (e.g., sarcasm). To address this, we implemented context-aware captioning by providing the preceding two segments as audio prompts (In-Context Learning). Specifically, for consecutive audio segments  $a_1$ ,  $a_2$ , and  $a_3$ , we calculate the conditional probability  $P(C_3|I, a_1, a_2)$  to generate the caption  $C_3$  for  $a_3$ .

## F Example

This section presents real-world podcast examples. Figure 7 visualizes the data processed by **Sommelier**, followed by an example of the corresponding JSON file.



**Figure 7** Visualization of preprocessing results for a 1-minute audio clip using a mel-spectrogram.

```

1 {
2   "metadata": {
3     "audio_duration_seconds": 120.0,
4     "audio_duration_minutes": 2.0,
5     "vad_sortformer": {
6       "processing_time_seconds": 0.9742708206176758,
7       "rt_factor": 0.008118923505147297
8     },
9     "whisper_large_v3": {
10      "processing_time_seconds": 14.903292655944824,
11      "rt_factor": 0.12419410546620686
12    },
13    "total_segments": 26,
14    "whisperx_alignment": {
15      "processing_time_seconds": 32.398998737335205,
16      "rt_factor": 0.26999165614446,
17      "enabled": true
18    },
19    "sepreformer_separation": {
20      "processing_time_seconds": 0.12217402458190918,
21      "rt_factor": 0.00101811687151591,
22      "overlap_threshold_seconds": 0.2,
23      "enabled": true
24    },
25    "flowse_denoising": {
26      "processing_time_seconds": 3.8639004230499268,
27      "rt_factor": 0.032199170192082724,
28      "enabled": true
29    }
30  },
31  "segments": [
32    {
33      "start": 0.0,
34      "end": 0.64,
35      "text": "Mr. Franklin?",
36      "text_whisper": "Mr. Franklin?",
37      "text_parakeet": "The Franklin?",
38      "text_canary": "The Franklin",
39      "speaker": "SPEAKER_00",
40      "language": "en",
41      "demucs": false,
42      "is_separated": true,
43      "sepreformer": false,
44      "words": [
45        {
46          "word": "Mr.",
47          "start": 0.0,
48          "end": 0.171,
49          "score": 0.414
50        },
51        {
52          "word": "Franklin?",
53          "start": 0.192,
54          "end": 0.661,
55          "score": 0.936
56        }
57      ]
58    },
59    {
60      "start": 0.48,
61      "end": 1.2,
62      "text": "I'm ready.",
63      "text_whisper": "I'm ready.",
64      "text_parakeet": "I'm ready.",
65      "text_canary": "I'm ready",
66      "speaker": "SPEAKER_01",
67      "language": "en",
68      "demucs": false,
69      "is_separated": true,
70      "sepreformer": false,
71      "words": [
72        {
73          "word": "I'm",
74          "start": 0.48,
75          "end": 0.861,
76          "score": 0.611
77        },
78        {
79          "word": "ready.",
80          "start": 0.9039999999999999,
81          "end": 1.221,

```



```

82         "score": 0.748
83     }
84 ]
85 },
86 {
87     "start": 1.12,
88     "end": 2.5599999999999996,
89     "text": "It's Ira Glass here",
90     "text_whisper": "Tyra Glass here.",
91     "text_parakeet": "It's Iraq Glass here.",
92     "text_canary": "It's Ira Glass here",
93     "speaker": "SPEAKER_00",
94     "language": "en",
95     "demucs": false,
96     "is_separated": true,
97     "sepreformer": false,
98     "words": [
99         {
100             "word": "Tyra",
101             "start": 1.12,
102             "end": 1.9220000000000002,
103             "score": 0.686
104         },
105         {
106             "word": "Glass",
107             "start": 1.943,
108             "end": 2.1900000000000004,
109             "score": 0.801
110         },
111         {
112             "word": "here.",
113             "start": 2.21,
114             "end": 2.5810000000000004,
115             "score": 0.75
116         }
117     ]
118 },
119 {
120     "start": 2.8,
121     "end": 7.279999999999999,
122     "text": "Oh you're the MC on the show I read about Oh great I read I read I read",
123     "text_whisper": "You're the emcee on the show, Ira. Oh, great. Ira, are you Ira? Ira?",
124     "text_parakeet": "Oh, you're the MC on the show, Ira. Oh, great. Ira Iron.",
125     "text_canary": "Oh you're the MC on the show I read about Oh great I read I read I read",
126     "speaker": "SPEAKER_01",
127     "language": "en",
128     "demucs": false,
129     "is_separated": true,
130     "sepreformer": true,
131     "words": [
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133             "word": "You're",
134             "start": 2.8,
135             "end": 3.2439999999999998,
136             "score": 0.498
137         },
138         {
139             "word": "the",
140             "start": 3.264,
141             "end": 3.3649999999999998,
142             "score": 0.685
143         },
144         {
145             "word": "emcee",
146             "start": 3.4459999999999997,
147             "end": 3.7479999999999998,
148             "score": 0.678
149         },
150         {
151             "word": "on",
152             "start": 3.7889999999999997,
153             "end": 3.8489999999999998,
154             "score": 0.932
155         },
156         {
157             "word": "the",
158             "start": 3.87,
159             "end": 3.9299999999999997,
160             "score": 0.977
161         },
162         {
163             "word": "show,"

```

```

164     "start": 3.9499999999999997,
165     "end": 4.213,
166     "score": 0.72
167 },
168 {
169     "word": "Ira.",
170     "start": 4.233,
171     "end": 5.282,
172     "score": 0.799
173 },
174 {
175     "word": "Oh,",
176     "start": 5.302,
177     "end": 5.484,
178     "score": 0.832
179 },
180 {
181     "word": "great.",
182     "start": 5.645,
183     "end": 5.968,
184     "score": 0.85
185 },
186 {
187     "word": "Ira,",
188     "start": 6.311,
189     "end": 6.614,
190     "score": 0.534
191 },
192 {
193     "word": "are",
194     "start": 6.634,
195     "end": 6.695,
196     "score": 0.211
197 },
198 {
199     "word": "you",
200     "start": 6.775,
201     "end": 6.936999999999999,
202     "score": 0.317
203 },
204 {
205     "word": "Ira?",
206     "start": 6.957,
207     "end": 7.119,
208     "score": 0.531
209 },
210 {
211     "word": "Ira?",
212     "start": 7.139,
213     "end": 7.3,
214     "score": 0.582
215 }
216 ],
217 "flowse_denoised": true
218 },
219 {
220     "start": 4.24,
221     "end": 5.76,
222     "text": "I am the MC on this show, yes.",
223     "text_whisper": "I am the MC on this show, yes.",
224     "text_parakeet": "I am the MC on the show, yes.",
225     "text_canary": "I am the MC on this show yes",
226     "speaker": "SPEAKER_00",
227     "language": "en",
228     "demucs": false,
229     "is_separated": true,
230     "sepreformer": true,
231     "words": [
232         {
233             "word": "I",
234             "start": 4.24,
235             "end": 4.61,
236             "score": 0.838
237         },
238         {
239             "word": "am",
240             "start": 4.63,
241             "end": 4.7330000000000005,
242             "score": 0.946
243         },
244         {
245             "word": "the",

```

```

246     "start": 4.774,
247     "end": 4.877000000000001,
248     "score": 0.215
249 },
250 {
251     "word": "MC",
252     "start": 4.897,
253     "end": 5.123,
254     "score": 0.683
255 },
256 {
257     "word": "on",
258     "start": 5.144,
259     "end": 5.205,
260     "score": 0.882
261 },
262 {
263     "word": "this",
264     "start": 5.226,
265     "end": 5.329000000000001,
266     "score": 0.287
267 },
268 {
269     "word": "show,",
270     "start": 5.349,
271     "end": 5.514,
272     "score": 0.455
273 },
274 {
275     "word": "yes.",
276     "start": 5.534000000000001,
277     "end": 5.781000000000001,
278     "score": 0.818
279 }
280 ],
281 "flowse_denoised": true
282 },
283 {
284     "start": 7.36,
285     "end": 8.48,
286     "text": "IRA. IRA.",
287     "text_whisper": "IRA. IRA.",
288     "text_parakeet": "Ira, IRA.",
289     "text_canary": "IRA I-R-A",
290     "speaker": "SPEAKER_00",
291     "language": "en",
292     "demucs": false,
293     "is_separated": true,
294     "sepreformer": false,
295     "words": [
296         {
297             "word": "IRA.",
298             "start": 7.36,
299             "end": 8.003,
300             "score": 0.779
301         },
302         {
303             "word": "IRA.",
304             "start": 8.024000000000001,
305             "end": 8.501000000000001,
306             "score": 0.648
307         }
308     ]
309 },
310 {
311     "start": 8.48,
312     "end": 11.2,
313     "text": "Oh, great. Now hold on one second Larry. Don't don't go away.",
314     "text_whisper": "Oh, great. Now, hold on one second there. Don't go away.",
315     "text_parakeet": "Oh, great. Now hold on one second Larry. Don't don't go away.",
316     "text_canary": "Oh great now hold on one second there don't go away",
317     "speaker": "SPEAKER_01",
318     "language": "en",
319     "demucs": false,
320     "is_separated": true,
321     "sepreformer": false,
322     "words": [
323         {
324             "word": "Oh,",
325             "start": 8.48,
326             "end": 8.825000000000001,
327             "score": 0.766

```

```

328     },
329     {
330         "word": "great.",
331         "start": 8.906,
332         "end": 9.292,
333         "score": 0.951
334     },
335     {
336         "word": "Now,",
337         "start": 9.678,
338         "end": 9.759,
339         "score": 0.146
340     },
341     {
342         "word": "hold",
343         "start": 9.779,
344         "end": 9.881,
345         "score": 0.504
346     },
347     {
348         "word": "on",
349         "start": 9.941,
350         "end": 10.002,
351         "score": 0.515
352     },
353     {
354         "word": "one",
355         "start": 10.063,
356         "end": 10.144,
357         "score": 0.348
358     },
359     {
360         "word": "second",
361         "start": 10.165000000000001,
362         "end": 10.327,
363         "score": 0.295
364     },
365     {
366         "word": "there.",
367         "start": 10.347000000000001,
368         "end": 10.693000000000001,
369         "score": 0.322
370     },
371     {
372         "word": "Don't",
373         "start": 10.733,
374         "end": 10.875,
375         "score": 0.69
376     },
377     {
378         "word": "go",
379         "start": 10.896,
380         "end": 10.997,
381         "score": 0.406
382     },
383     {
384         "word": "away.",
385         "start": 11.017,
386         "end": 11.22,
387         "score": 0.513
388     }
389 ]
390 },
391 {
392     "start": 12.4,
393     "end": 13.44,
394     "text": "Hello?",
395     "text_whisper": "Hello?",
396     "text_parakeet": "Hello?",
397     "text_canary": "Hello",
398     "speaker": "SPEAKER_01",
399     "language": "en",
400     "demucs": false,
401     "is_separated": true,
402     "sepreformer": false,
403     "words": [
404         {
405             "word": "Hello?",
406             "start": 12.4,
407             "end": 13.461,
408             "score": 0.415
409         }

```

```

410 | ]
411 | },
412 | {
413 |     "start": 14.48,
414 |     "end": 18.080000000000002,
415 |     "text": "Sheldon, call me after 3 o'clock. I've got great news for you. Ira...",
416 |     "text_whisper": "Sheldon, call me after 3 o'clock. I've got great news for you. Ira...",
417 |     "text_parakeet": "Sheldon, call me after three o'clock. I've got great news for you. Ira.",
418 |     "text_canary": "Shelton McCoomy at three o'clock got great news for you. Irum",
419 |     "speaker": "SPEAKER_01",
420 |     "language": "en",
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428 |             "end": 15.046000000000001,
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433 |             "start": 15.067,
434 |             "end": 15.208,
435 |             "score": 0.237
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444 |             "word": "after",
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457 |             "start": 15.774000000000001,
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494     "end": 17.008,
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508     "text": "Yes.",
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510     "text_parakeet": "Yeah.",
511     "text_canary": "Yeah",
512     "speaker": "SPEAKER_00",
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522             "score": 0.471
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527     "start": 18.96,
528     "end": 33.28,
529     "text": "So uh listen, Tony if the phone rings, take it in the back and then tell me then come out and tell me who it is. is.
        Just Just say Joe's being with a camera crew. Just for about 10 minutes. We'll do about five minutes, ten minutes,
        right, Irv?",
530     "text_whisper": "So, listen, Tony, if the phone rings, take it in the back, and then come out and tell me who it is. Just say
        Joe's being with a camera crew. Just for about ten minutes. We'll do about five minutes, ten minutes, right, Iris?",
531     "text_parakeet": "So listen, Tony. If the phone rings, take it in the back and tell me, then come out and tell me who it is.
        Just say Joe's being with a camera crew. Just for about 10 minutes. We'll do about five minutes, ten minutes, right,
        Ivory?",
532     "text_canary": "So uh listen Tony if the phone rings take it in the back and then tell me then come out and tell me who it is
        just say just say Joe's being with the camera crew just for about 10 minutes we'll do a five minute ten minutes right
        Irv?",
533     "speaker": "SPEAKER_01",
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553             "start": 20.946,
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558             "word": "if",
559             "start": 21.928,
560             "end": 22.029,
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563         {
564             "word": "the",
565             "start": 22.069000000000003,
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646   },
647   {
648     "word": "me",

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675     "score": 0.927
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798         "word": "Iris?",
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810     "text_parakeet": "Yep.",
811     "text_canary": "That's fair",
812     "speaker": "SPEAKER_00",

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858             "word": "know",
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875     "text_whisper": "Great thing about starting a new show is utter anonymity. Nobody really knows what to expect from you.",
876     "text_parakeet": "Great thing about starting a new show is utter anonymity. Nobody who really knows what to expect from you.",
877     "text_canary": "The great thing about starting a new show is utter anonymity. Nobody really knows what to expect from you.",
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969     "word": "to",
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971     "end": 47.871,
972     "score": 0.827
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974 {
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988     "start": 49.459,
989     "end": 49.7,
990     "score": 0.861
991 }
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993 },
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995     "start": 50.64,
996     "end": 53.04,
997     "text": "This interviewee did not know us from Adam.",
998     "text_whisper": "This interviewee did not know us from Adam.",
999     "text_parakeet": "This interviewee did not know us from Adam.",
1000     "text_canary": "This interviewee did not know us from Adam",
1001     "speaker": "SPEAKER_02",
1002     "language": "en",
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1059 "end": 63.04,
1060 "text": "Okay, well what? About a minute. We're one minute five into the new show right now. it is stretching in front of us.
      The perfect future.",
1061 "text_whisper": "Okay, well, what? About a minute. We're one minute five into the new show right now. It is stretching in
      front of us. The perfect future.",
1062 "text_parakeet": "Okay, well what? About a minute. We're one minute five into the new show. Right now, it is stretching in
      front of us. A perfect future.",
1063 "text_canary": "Okay well what about a minute well one minute five into the new show right now it is stretching in front of
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1064 "speaker": "SPEAKER_02",
1065 "language": "en",
1066 "demucs": true,
1067 "is_separated": true,
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1127     "end": 57.774,
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1131     "word": "into",
1132     "start": 57.814,
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1134     "score": 0.774
1135   },

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1161 |   "word": "now.",
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1167 | }
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1169 | }

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